

TUGAS MANDIRI MODUL 13

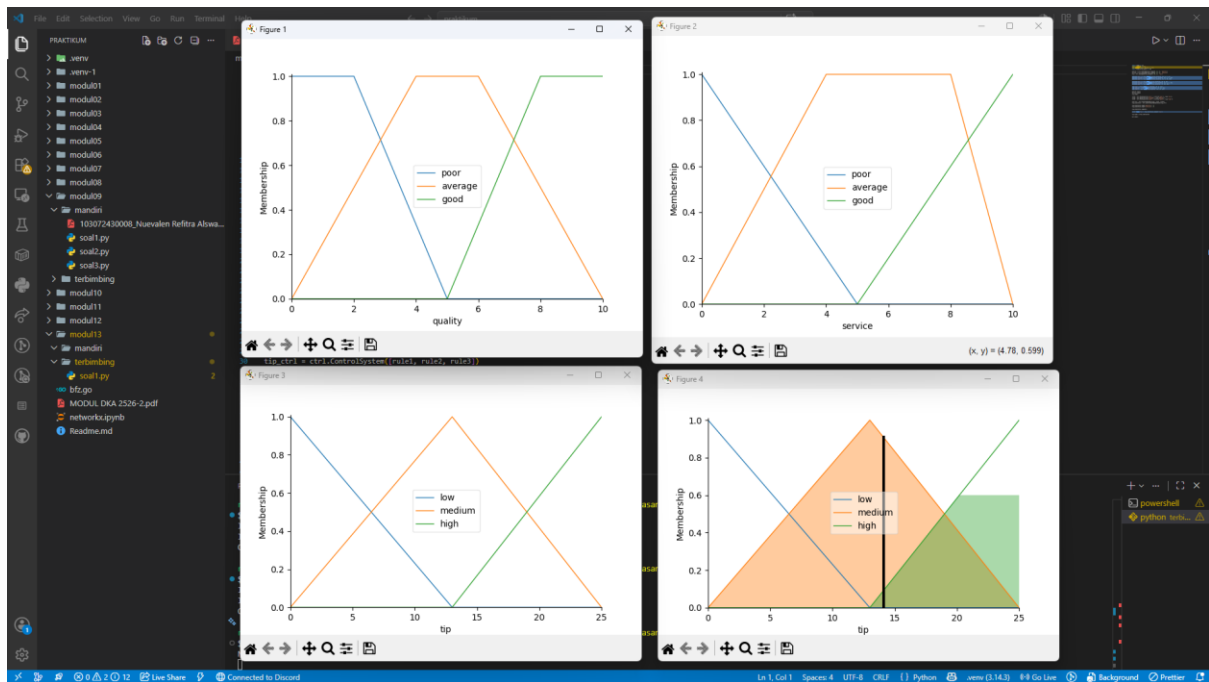
Scikit-Fuzzy

NAMA : Nuevalen Refitra Alswando
NIM : 10307243008
KELAS : IF-04-01

Code

```
modul13 > terbimbing >  soal1.py > ...
1  import numpy as np
2  import skfuzzy as fuzz
3  from skfuzzy import control as ctrl
4  import matplotlib.pyplot as plt
5
6  quality = ctrl.Antecedent(np.arange(0, 11, 1), 'quality')
7  service = ctrl.Antecedent(np.arange(0, 11, 1), 'service')
8  tip = ctrl.Consequent(np.arange(0, 26, 1), 'tip')
9
10 quality['poor'] = fuzz.trapmf(quality.universe, [0, 0, 2, 5])
11 quality['average'] = fuzz.trapmf(quality.universe, [0, 4, 6, 10])
12 quality['good'] = fuzz.trapmf(quality.universe, [5, 8, 10, 10])
13
14 service['poor'] = fuzz.trimf(service.universe, [0, 0, 5])
15 service['average'] = fuzz.trapmf(service.universe, [0, 4, 8, 10])
16 service['good'] = fuzz.trimf(service.universe, [5, 10, 10])
17
18 tip['low'] = fuzz.trimf(tip.universe, [0, 0, 13])
19 tip['medium'] = fuzz.trimf(tip.universe, [0, 13, 25])
20 tip['high'] = fuzz.trimf(tip.universe, [13, 25, 25])
21
22 quality.view()
23 service.view()
24 tip.view()
25
26 rule1 = ctrl.Rule(quality['poor'] | service['poor'], tip['low'])
27 rule2 = ctrl.Rule(service['average'], tip['medium'])
28 rule3 = ctrl.Rule(quality['good'] | service['good'], tip['high'])
29
30 tip_ctrl = ctrl.ControlSystem([rule1, rule2, rule3])
31 tip_simulation = ctrl.ControlSystemSimulation(tip_ctrl)
32
33 tip_simulation.input['quality'] = 6
34 tip_simulation.input['service'] = 8
35
36 tip_simulation.compute()
37
38 print("Nilai Tip:", tip_simulation.output['tip'])
39
40 tip.view(sim=tip_simulation)
41
42 plt.show()
```

Output



Penjelasan

Pada codingan tersebut, saya membuat sistem fuzzy menggunakan library Scikit-Fuzzy dengan tiga variabel utama yaitu quality, service, dan tip. Saya mendefinisikan fungsi keanggotaan untuk quality dan service sesuai grafik pada soal menggunakan fungsi trapmf dan trimf, kemudian membuat fungsi output tip dengan kategori low, medium, dan high. Setelah itu, saya membuat aturan fuzzy untuk menentukan besar tip berdasarkan kualitas dan pelayanan. Selanjutnya saya memasukkan nilai input, melakukan proses inferensi fuzzy menggunakan compute(), lalu menampilkan hasil output tip beserta visualisasi grafik fungsi keanggotaannya.